adapted on
Lynford L. Goddard
Brian T. Cunningham

Lectures 4

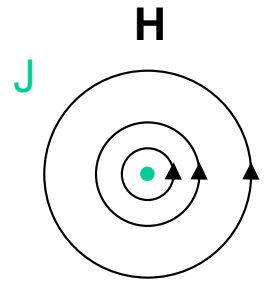
Divergence and Curl

Maxwell's Equations in Differential Form

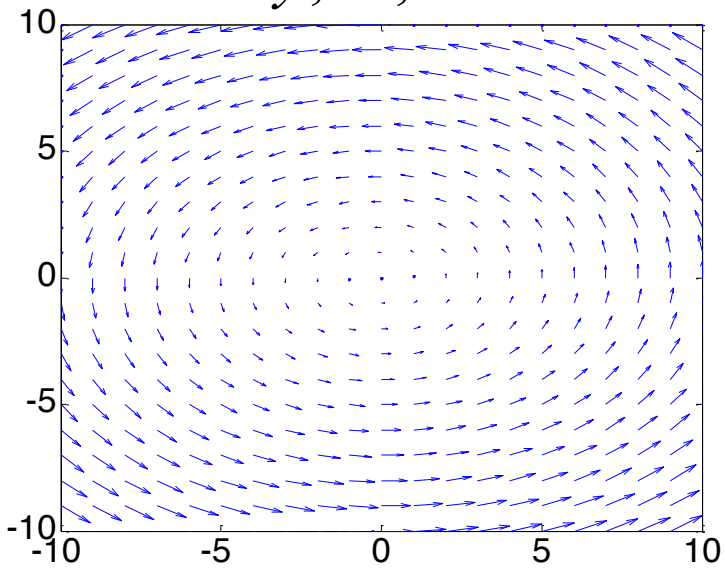
Victoria Shao
yangshao@Illinois.edu

© Lynford Goddard and Brian Cunningham - Copying, publishing or distributing without express written permission is prohibited.

Curl



$$\vec{v} = < -3y, 2x, 0 >$$



$$\nabla \times \vec{v} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix} = 5\hat{a}_z$$

$$= \hat{a}_x \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right)$$

$$-\hat{a}_y \left(\frac{\partial v_z}{\partial x} - \frac{\partial v_x}{\partial z} \right)$$

$$+\hat{a}_z \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right)$$

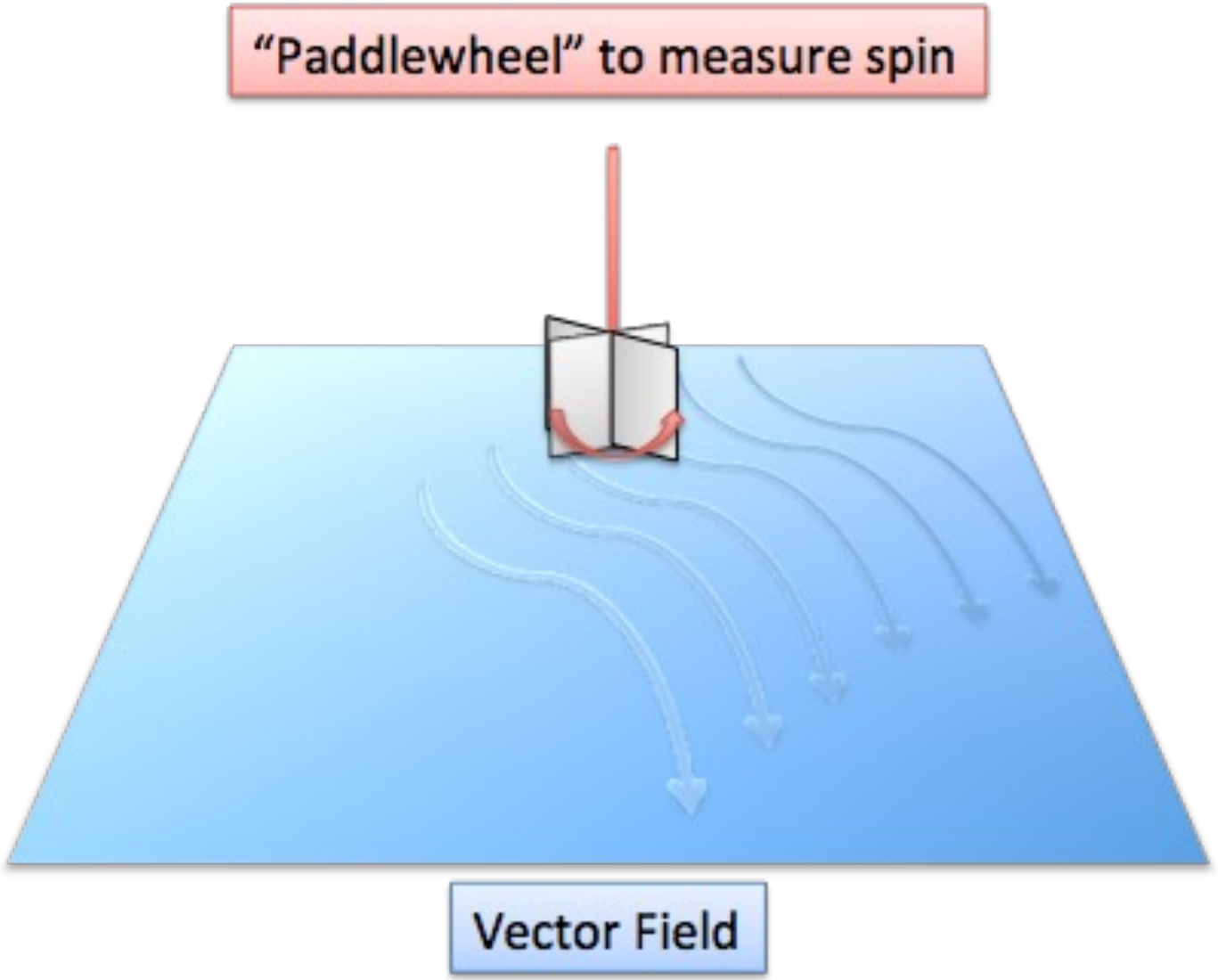
circulation per unit area

$\nabla \times \vec{v}$ is a measure of how much the vector field $\vec{v}$ circulates at a given point

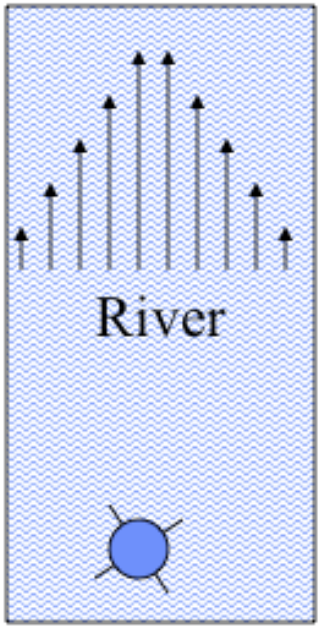
A place of high curl is like a whirlpool

– Everywhere here is a whirlpool of strength 5

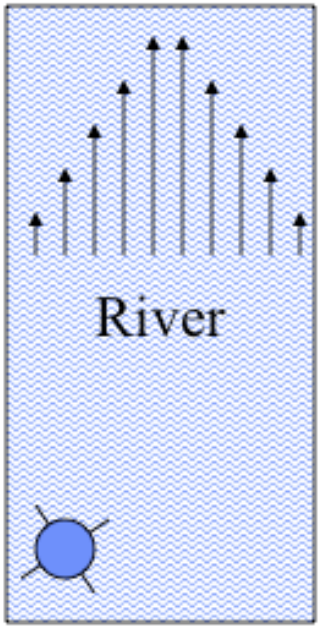
Physical Interpretation of Curl



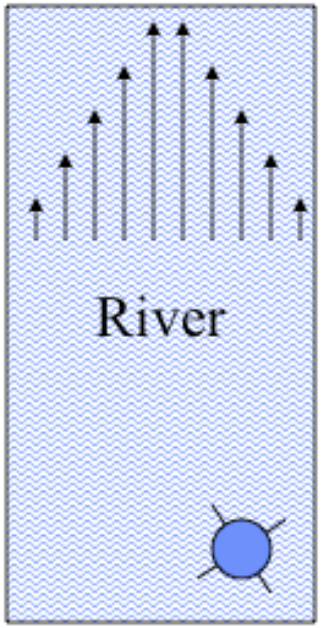
Physical Interpretation of Curl



No rotation!



Anti-clockwise rotation.

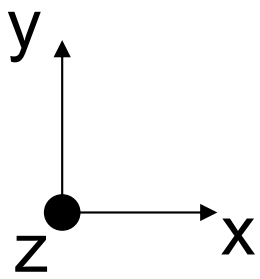


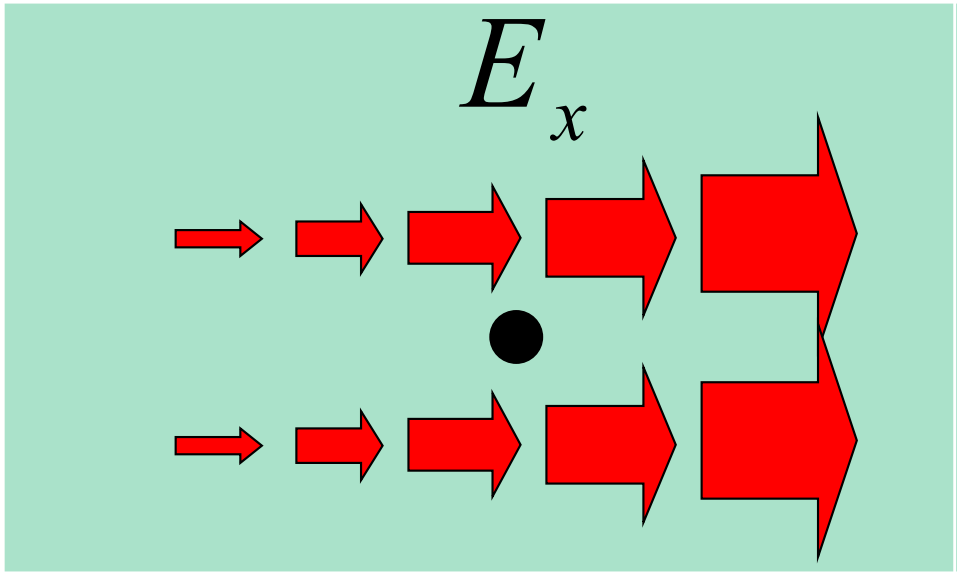
Clockwise rotation.

Meaning of Curl

- The curl meter only spins if there is a non-uniformity in the vector field in a direction **perpendicular** to the field
  - Curl describes variation ACROSS the flow of the field
- Rotation rate is proportional to the degree of non-uniformity
- Rotation is described with a magnitude and a direction – so it's a VECTOR and it's given by the right hand rule

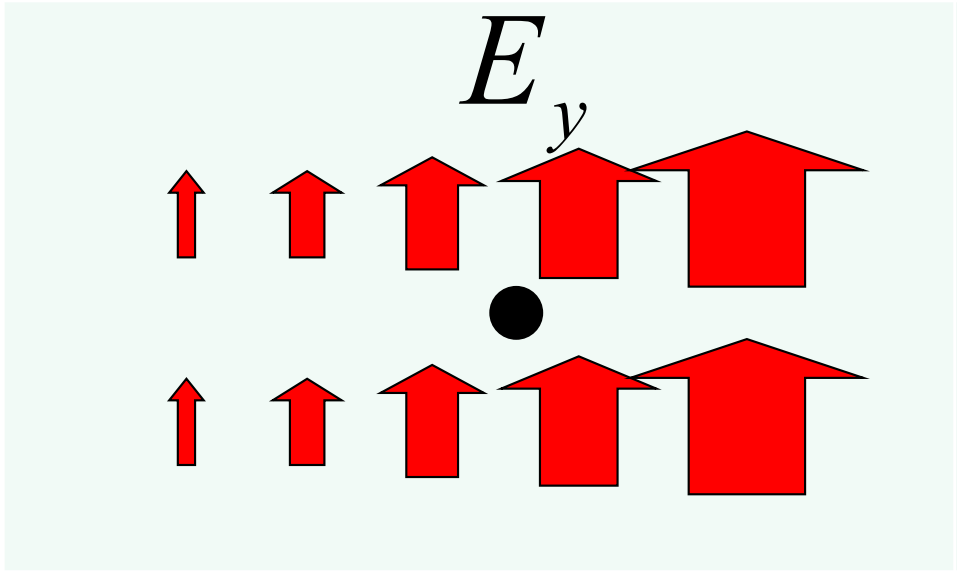
Curl





$$\nabla \times \vec{E} = 0$$

True or false?

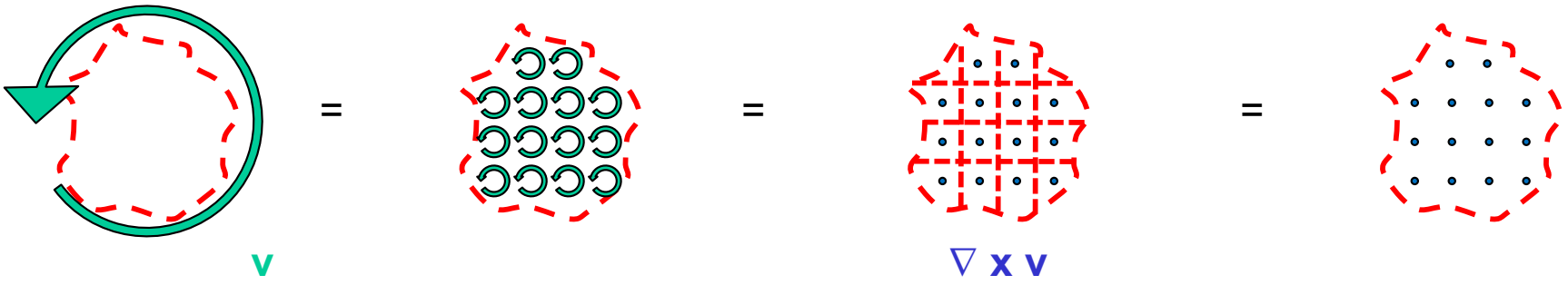


$$\nabla \times \vec{E} = 0$$

True or false?

Stokes' Theorem

$$\oint_C \vec{v} \bullet d\vec{l} = \sum \oint_{C_i} \vec{v} \bullet d\vec{l} = \sum \iint_{S_i} (\nabla \times \vec{v}) \bullet d\vec{S} = \iint_S (\nabla \times \vec{v}) \bullet d\vec{S}$$



∇ x **v** is a measure of how much the vector field **v** circulates at a given point

A place of high curl is like a whirlpool

Thus, adding up the circulation from each whirlpool inside a region = the total circ. along the boundary

v is a conservative field iff ∇ x **v** = 0

$$\oint_C \vec{v} \bullet d\vec{l} = 0 = \iint_S (\nabla \times \vec{v}) \bullet d\vec{S}$$

The following are therefore equivalent:

v is conservative

$$\oint_C \vec{v} \bullet d\vec{l} = 0$$

v is curl-free

$$\int_a^b \vec{v} \bullet d\vec{l} \text{ is path independent}$$

$$\vec{v} = -\nabla f \text{ for some scalar potential f}$$

Example: Curl Free Field

Recall that the static field of a line charge λ distributed on the z-axis is

$$\mathbf{E}(x,y,z) = \hat{r}\frac{\lambda}{2\pi\epsilon_o r},$$

where $r^2 = x^2 + y^2$ and $\hat{r} = \hat{x}\cos\phi + \hat{y}\sin\phi = (\frac{x}{r}, \frac{y}{r}, 0).$

Show that field $\vec{E}$ is curl free.

Review at home
Prof. Kudeki's Lect04.pdf
Example 4

Example: Curl Free Field

Recall that the static field of a line charge λ distributed on the z-axis is

$$\mathbf{E}(x,y,z) = \hat{r}\frac{\lambda}{2\pi\epsilon_o r},$$

where $r^2 = x^2 + y^2$ and $\hat{r} = \hat{x}\cos\phi + \hat{y}\sin\phi = (\frac{x}{r}, \frac{y}{r}, 0).$

Show that field $\vec{E}$ is curl free.

Solution:

$$\mathbf{E} = \frac{\lambda}{2\pi\epsilon_o}(\frac{x}{r^2}, \frac{y}{r^2}, 0)$$

$$\nabla \times \mathbf{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix} = \frac{\lambda}{2\pi\epsilon_o} \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{x}{r^2} & \frac{y}{r^2} & 0 \end{vmatrix}$$

$$= \frac{\lambda}{2\pi\epsilon_o}\hat{z}(\frac{\partial}{\partial x}\frac{y}{r^2} - \frac{\partial}{\partial y}\frac{x}{r^2})$$

Example: Curl Free Field

Recall that the static field of a line charge λ distributed on the z-axis is

$$\mathbf{E}(x,y,z) = \hat{r}\frac{\lambda}{2\pi\epsilon_o r},$$

where $r^2 = x^2 + y^2$ and $\hat{r} = \hat{x}\cos\phi + \hat{y}\sin\phi = (\frac{x}{r}, \frac{y}{r}, 0).$

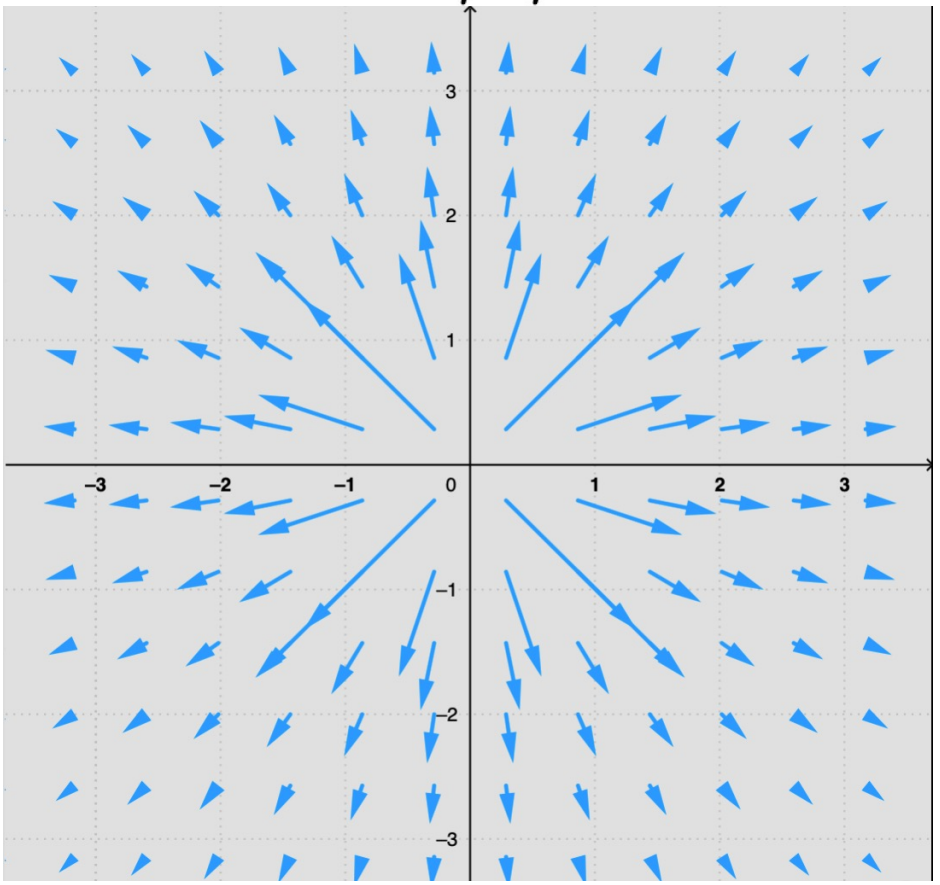
Show that field $\vec{E}$ is curl free.

Solution:

$$\frac{\partial}{\partial x}\frac{y}{r^2} - \frac{\partial}{\partial y}\frac{x}{r^2} = y\frac{\partial}{\partial x}\frac{1}{r^2} - x\frac{\partial}{\partial y}\frac{1}{r^2}$$

$$= y\frac{-2x}{r^4} - x\frac{-2y}{r^4} = 0$$

So $\nabla\times\vec{E} = \mathbf{0}$



Maxwell's Equations in Differential Form

Faraday's Law

$$\oint_C \vec{E} \bullet d\vec{l} = -\frac{d}{dt}\iint_S \vec{B} \bullet d\vec{S}$$



Stokes' Thm.

$$\oint_C \vec{E} \bullet d\vec{l} = \iint_S (\nabla \times \vec{E}) \bullet d\vec{S} = -\frac{d}{dt}\iint_S \vec{B} \bullet d\vec{S}$$

$$\Rightarrow \boxed{\nabla \times \vec{E} = -\frac{d\vec{B}}{dt}}$$



Maxwell's Equations in Differential Form

Ampere's Law

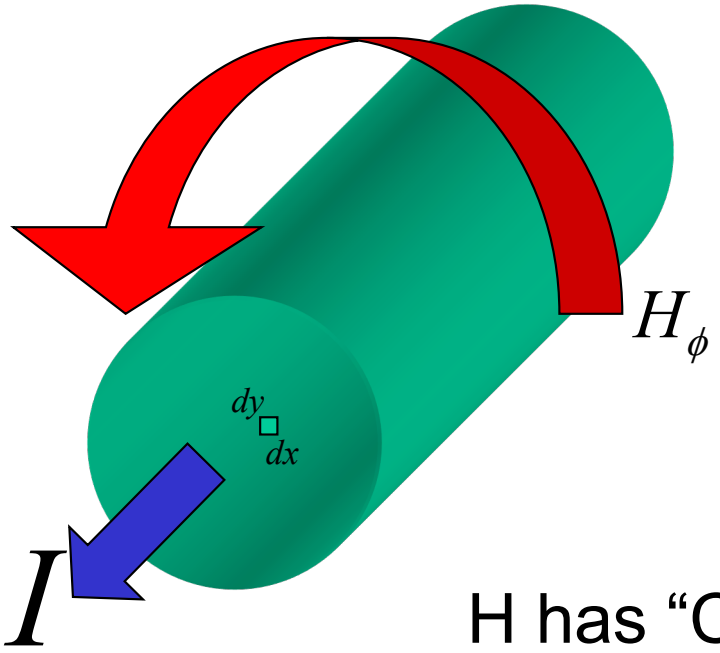
$$\oint_C \vec{H} \bullet d\vec{l} = \iint_S \vec{J} \bullet d\vec{S} + \frac{d}{dt} \iint_S \vec{D} \bullet d\vec{S}$$

Stokes' Thm

$$\oint_C \vec{H} \bullet d\vec{l} = \iint_S (\nabla \times \vec{H}) \bullet d\vec{S} = \iint_S (\vec{J} + \frac{d\vec{D}}{dt}) \bullet d\vec{S}$$

$$\Rightarrow \boxed{\nabla \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}}$$

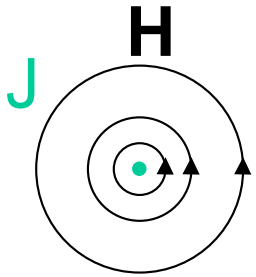
Ampere's Law In Differential Form



$$MMF = \oint \vec{H} \bullet d\vec{l} \neq 0$$

If there is any current going through a particular point

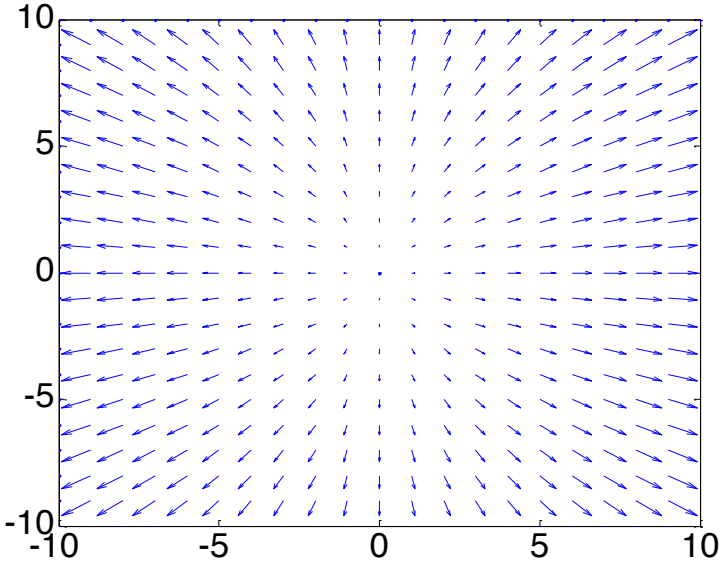
H has "CURL" at a point if there is current going through the point:

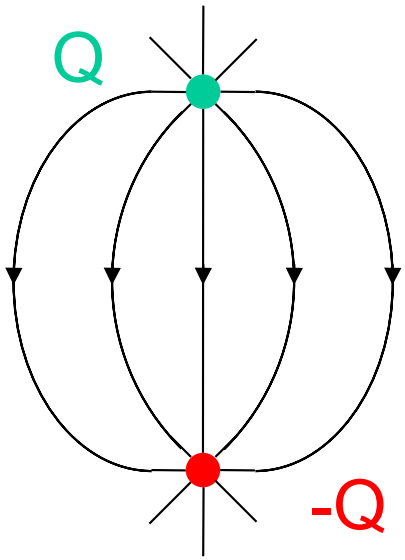


CONDUCTION current
DISPLACEMENT current

Divergence

$$\vec{v} = \langle 3x, 2y, 0 \rangle$$





$$\nabla \bullet \vec{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 5$$

flux per unit volume

$\nabla \cdot \vec{v}$ is a measure of how many field lines for **v** are created at a given point

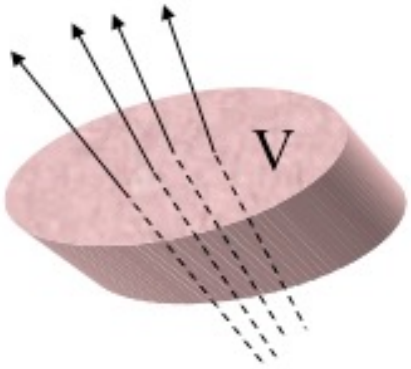
A place of high divergence is like a water faucet

— Everywhere here is a faucet of strength 5

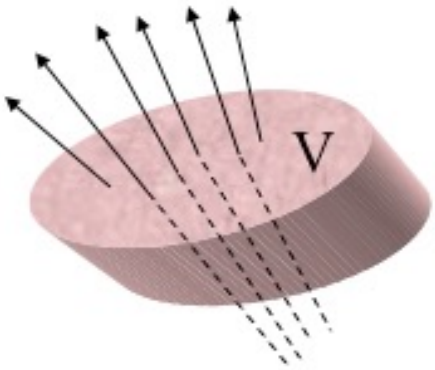
$$\text{Divergence} = \lim_{\text{Vol}\to 0} \frac{\text{Flux}}{\text{Vol}}$$

Meaning of Divergence

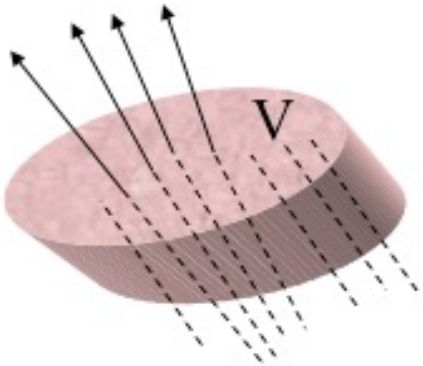
$$\text{Divergence} = \lim_{V \to 0} \frac{\text{Flux}}{V}$$



Flux in = flux out so no sources or sinks inside V.



Flux out > flux in Positive divergence. Must be a source inside V.

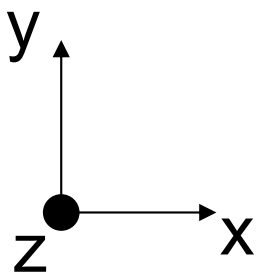


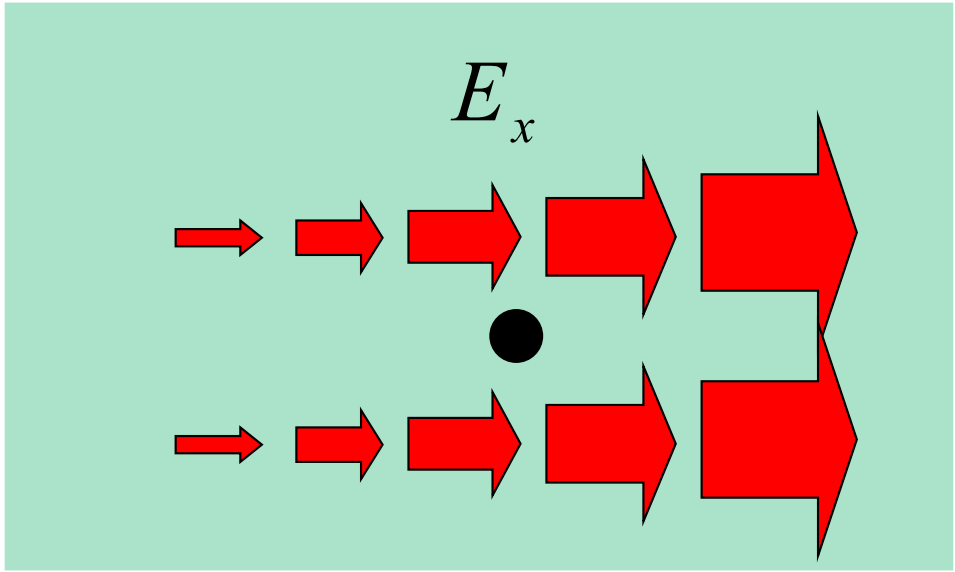
Flux out < flux in Negative divergence. Must be a sink or drain inside V.

Meaning of Divergence

- There is divergence if there is a non-uniformity in the vector field in a direction **parallel** to the field
  - Divergence describes variation ALONG the flow of the field
- Divergence is proportional to the degree of non-uniformity
- Divergence is described only by the magnitude – so it's a SCALAR

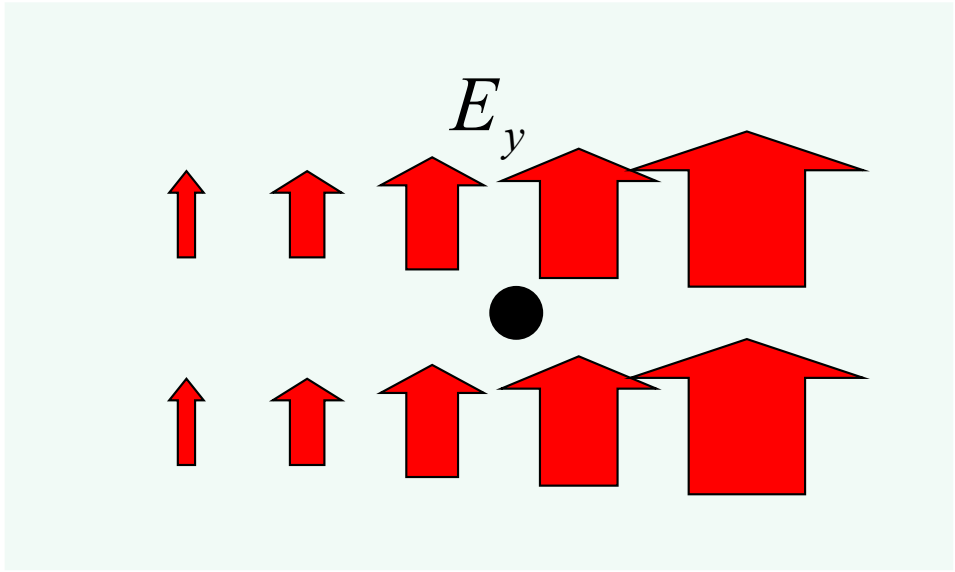
Divergence





$$\nabla \cdot \vec{E} = 0$$

True or false?



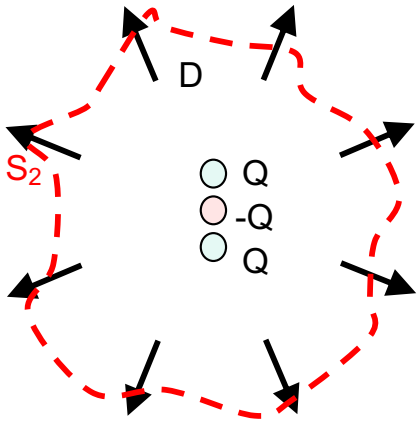
$$\nabla \cdot \vec{E} = 0$$

True or false?

Example: Divergence

- Find the divergence of $\vec{A} = (x-2)^2\hat{x} + 2\sin z\,\hat{y}$

Divergence Theorem



$$\int_V \nabla \cdot \vec{\boldsymbol{D}}\, dV = \oint_S \vec{\boldsymbol{D}} \cdot d\vec{\boldsymbol{S}}$$

For a general vector field $\vec{\boldsymbol{v}}$

$$\iiint_V \nabla \bullet \vec{v}\; dV = \oiint_{\partial V} \vec{v} \bullet d\vec{S}$$

$\nabla \cdot \vec{\boldsymbol{v}}$ is a measure of how many field lines for $\vec{\boldsymbol{v}}$ are created at a given point

Thus, adding up the net # lines created inside a volume = the flux of lines out its boundary

Maxwell's Equations in Differential Form

Gauss' Law

$$\oiint_S \vec{B} \bullet d\vec{S} = 0$$

Divergence Thm

$$\oiint_S \vec{B} \bullet d\vec{S} = \iiint_V \nabla \bullet \vec{B}\ dV = 0$$

$$\Rightarrow \boxed{\nabla \bullet \vec{B} = 0}$$

Differential form of Gauss's law for magnetism

Maxwell's Equations in Differential Form

Gauss' Law

$$\oiint_S \vec{D} \bullet d\vec{S} = \iiint_V \rho dV$$

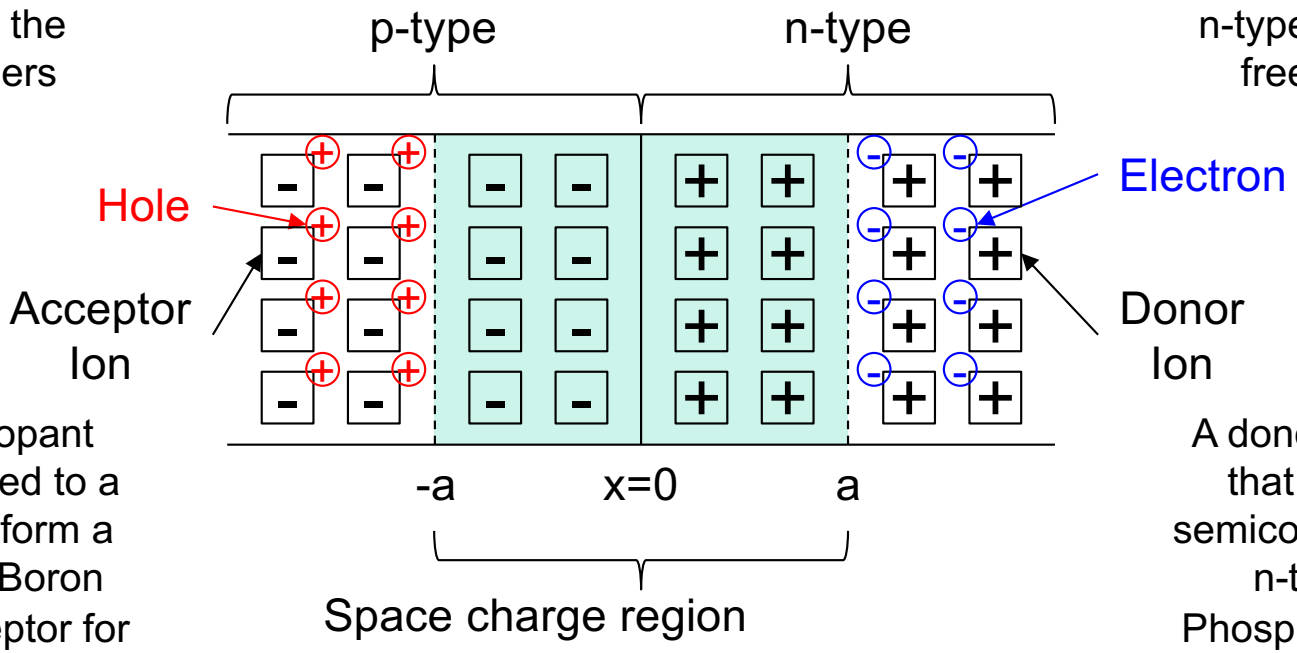
Divergence Thm

$$\oiint_S \vec{D} \bullet d\vec{S} = \iiint_V \nabla \bullet \vec{D}\ dV = \iiint_V \rho dV$$

$$\Rightarrow\ \nabla \bullet \vec{D} = \rho$$

E-field of pn junction (ECE 340)

p-type: holes are the free charge carriers



n-type: electrons are the free charge carriers

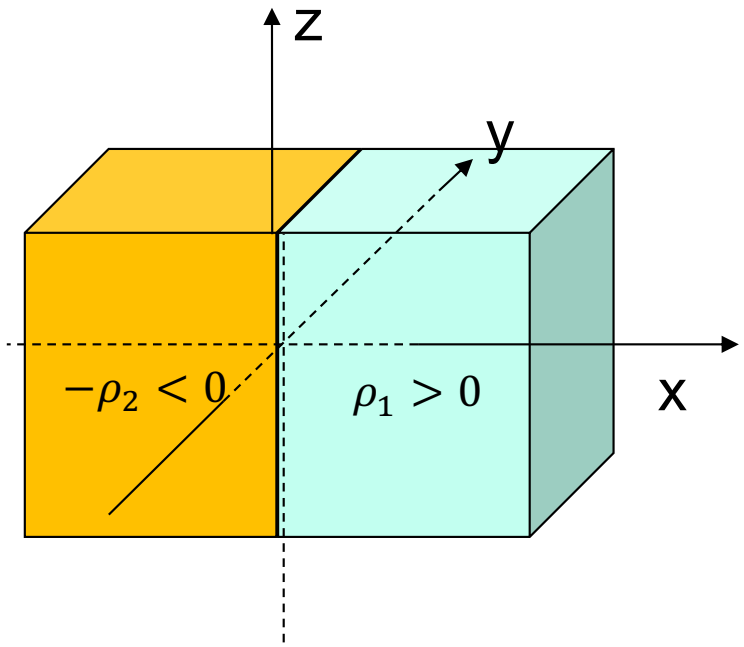
An acceptor is a dopant atom that when added to a semiconductor can form a p-type region, e.g. Boron (group III) is an acceptor for Silicon (group IV)

A donor is a dopant atom that when added to a semiconductor can form an n-type region, e.g. Phosphorus (group V) is a donor for Silicon (group IV)

- Find the **E**-field for an evenly doped: $N_d=N_a$ pn junction: $\rho=\{ -\rho_0=-qN_a$ for $-a<x<0$,
  $\{ \rho_0=qN_d$ for $0<x<a$,
  $\{ 0$ for $|x|>a$

From Example 3.5 (p 145) of old book

Example: **E** due to slabs of charge



Volume charge density (C/m^3)

Two infinite slabs have equal but opposite amount of total charge are located $-W_1 < x < W_2$. If we have $\rho_1 W_1 = \rho_2 W_2$, find **E**.

Review at home
Prof. Kudeki's Lect04.pdf
Example 2

Continuity Equation in Differential Form

Continuity Eqn

$$\oiint_S \vec{J} \bullet d\vec{S} = -\frac{d}{dt}\iiint_V \rho dV$$

Divergence Thm

$$\oiint_S \vec{J} \bullet d\vec{S} = \iiint_V \nabla \bullet \vec{J}\ dV = -\frac{d}{dt}\iiint_V \rho dV$$

$$\Rightarrow\ \nabla \bullet \vec{J} = -\frac{d\rho}{dt}$$

Useful Relationships

$$\nabla \bullet (\nabla \times \vec{A}) = 0$$

$$\nabla \times (\nabla f) = 0 \quad \text{We already knew } \nabla f \text{ is conservative}$$

$$\nabla \times (\nabla \times \vec{A}) = \nabla(\nabla \bullet \vec{A}) - \nabla^2 \vec{A} \quad \text{Must use cartesian coordinates!}$$

$$\nabla^2 \vec{A} \equiv (\nabla^2 A_x)\hat{x} + (\nabla^2 A_y)\hat{y} + (\nabla^2 A_z)\hat{z}$$

$$0 = \nabla \bullet (\nabla \times \vec{H}) = \nabla \bullet (\vec{J} + \frac{d\vec{D}}{dt})$$

$$= \nabla \bullet \vec{J} + \frac{d}{dt}(\nabla \bullet \vec{D}) = \nabla \bullet \vec{J} + \frac{d\rho}{dt}$$

$$\therefore \nabla \bullet \vec{J} = -\frac{d\rho}{dt}$$

The continuty eqn. is contained in Maxwell's Eqns.

Example: Continuity Eqn.

- For $\mathbf{J}=J_0(x^2\mathbf{a}_x+y^2\mathbf{a}_y+z^2\mathbf{a}_z)$, find $d\rho/dt$ at the point (0.02,0.01,0.01)

From D3.6a (p 149) of old book

Summary of Maxwell's Equations

Faraday's Law $\oint_C \vec{E} \bullet d\vec{l} = -\frac{d}{dt}\iint_S \vec{B} \bullet d\vec{S}$ $\quad \nabla \times \vec{E} = -\frac{d\vec{B}}{dt}$

Ampere's Law $\oint_C \vec{H} \bullet d\vec{l} = \iint_S \vec{J} \bullet d\vec{S} + \frac{d}{dt}\iint_S \vec{D} \bullet d\vec{S}$ $\quad \nabla \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}$

Gauss' Law $\oiint_S \vec{B} \bullet d\vec{S} = 0$ $\quad \nabla \bullet \vec{B} = 0$

Gauss' Law $\oiint_S \vec{D} \bullet d\vec{S} = \iiint_V \rho dV$ $\quad \nabla \bullet \vec{D} = \rho$

Continuity Eq. $\oiint_S \vec{J} \bullet d\vec{S} = -\frac{d}{dt}\iiint_V \rho dV$ $\quad \nabla \bullet \vec{J} = -\frac{d\rho}{dt}$